Program: **Diploma in Food Processing Technology**	
Course Code: **6427**	Course Title: **Fruit and Vegetable Processing Lab**
Semester: **6**	Credits: **1.5**
Course Category: **Program core**	
Periods per week: **3(L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- Determine various processing and preservation techniques for fruits and vegetables
- Study the biochemical changes of the food materials during various processing techniques

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Awareness about processing techniques of fruits and vegetables		Fruits and vegetable processing	6

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply techniques in the preparation of various fruit-based products, including jelly, fruit leather, gummies, fruit syrup, osmotic dehydration of carrot and grapes, and the preparation of amla preserve and dried fruit products like aam papad and bars.	24	Applying
CO2	Apply techniques for processing vegetables by drying and preparing fruit cordials according to FPO specifications.	7	Applying
CO3	Apply techniques for processing and preservation of peas through bottling and blanching, while assessing the adequacy of the blanching process.	6	Applying

CO4	Apply methods for controlling enzymatic browning in fruits and vegetables.	8	Applying

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Blooms Taxonomy Level
CO1	**Apply techniques in the preparation of various fruit-based products, including jelly, fruit leather, gummies, fruit syrup, osmotic dehydration of carrot and grapes, and the preparation of amla preserve and dried fruit products like aam papad and bars.**		
M1.01	Demonstrate the ability to prepare jelly by following the appropriate steps and techniques for mixing, dissolving, and setting the ingredients.	3	Understanding
M1.02	Apply the techniques for preparing fruit leather by selecting, processing, and dehydrating fruits to produce a finished product.	6	Applying
M1.03	Demonstrate the preparation of gummies by applying the necessary techniques and methods, using appropriate ingredients, tools, and equipment to create a high-quality product	3	Applying
M1.04	Demonstrate the ability to prepare fruit syrup by applying appropriate techniques and processes to combine fruits, sugar, and water, ensuring consistency and flavor balance.	3	Applying
M1.05	Apply the principles of osmotic dehydration to process carrots, demonstrating the ability to select appropriate parameters and techniques for effective moisture reduction.	3	Applying

M1.06	Apply the principles of osmotic dehydration to effectively dehydrate grapes, utilizing appropriate techniques and methods to enhance the preservation and quality of the final product.	3	Applying
M1.07	Apply knowledge and skills to prepare amla preserve and dried fruit products, including aam papad and bars, using appropriate methods and techniques.	3	Applying
CO2	**Apply techniques for processing vegetables by drying and preparing fruit cordials according to FPO specifications.**		
M2.01	Apply the techniques of drying to process various vegetables, demonstrating the ability to execute appropriate methods for preservation and assess the impact on the quality of the produce.	2	Applying
M2.02	Apply the FPO specifications to prepare fruit cordials, demonstrating the appropriate techniques and processes.	3	Applying
	Lab Test –I	2	
CO3	**Apply techniques for processing and preservation of peas through bottling and blanching, while assessing the adequacy of the blanching process.**		
M3.01	Apply techniques for processing and preserving peas by utilizing high temperatures in the bottling method.	3	Applying
M3.02	Apply appropriate techniques for blanching a given sample (pea) and assess the adequacy of the process.	3	Applying
CO4	**Apply methods for controlling enzymatic browning in fruits and vegetables.**		
M4.01	Apply the principles of enzymatic browning in fruits and demonstrate effective methods to control the browning process.	3	Applying
M4.02	Apply knowledge of enzymatic browning in vegetables and demonstrate control techniques to minimize its occurrence during food processing and storage.	3	Applying
	Lab Test– II	2	

Text/Reference

T/R	Book Title / Author
R1	Fellows, P J. Food Processing Technology: Principles and Practice. 2nd Edition, CRC/ Woodhead, 1997. ISBN 0 8493 0887 9, ISBN 1 85573 533 4
R2	Sivasankar, B. Food Processing and Preservation, Prentice Hall of India, 2002. ISBN 10: 8120320867ISBN 13: 9788120320864
R3	Salunke, D. K and S. S Kadam Hand Book of Fruit Science and Technology: Production, Composition, Storage and Processing. Marcel Dekker, 1995, ISBN 0 8247 9643 8

Online Resources

Sl.No	Website Link
1.	https://www.appropedia.org/Toddy_and_Palm_Wine_(Practical_Action_Technical_Brief)
2.	https://www.bhu.ac.in/Content/Syllabus/Syllabus_3006309920200412060036.pdf
3.	https://vikaspedia.in/agriculture/post-harvest-technologies/technologies-for-agri-horti-crops/technologies-for-ripening-fruits